

Systematic literature review on tail risk in financial markets: A Pre-register report

Abstract

This paper provides a pre-registered research plan for a systematic literature review (SLR) of tail risk. Tail risk is the risk of extreme, low-probability losses in financial markets, and such events can abruptly reprice assets and destabilize the wider financial system. However, research on tail risk is dispersed across risk measures, asset classes, and countries. This pre-registered SLR aims to organize that fragmented landscape using a transparent, replicable protocol. Following PRISMA principles, we will extract studies from major academic databases and using a pre-specified code book to classify them. Planned outputs include a taxonomy of tail-risk measures, synthesis tables and mechanism maps, bibliometric summaries where feasible, and a structured agenda for future research.

Keywords: Tail risk; Financial Markets; Systematic Literature Review (SLR).

"You don't know who's swimming naked until the tide goes out." –Warren Buffet

1 Introduction

On October 10, 2025, global financial markets reeled when the U.S. President Donald Trump announced an abrupt 100% additional tariff on Chinese imports, a move that erased more than 19 billion USD in cryptocurrency positions within hours and triggered sharp stock market sell-offs across multiple sectors. This sudden escalation exposed the fragility of markets that often appear stable under normal conditions, revealing how hidden vulnerabilities can erupt into widespread turmoil.¹ This event reflects the broader nature of tail risk: low-probability but high-impact occurrences capable of destabilizing markets and economies. Episodes such as the 2008 Global Financial Crisis (GFC), the COVID-19 pandemic, and the Russia–Ukraine war (2022) have similarly highlighted the importance of understanding how extreme risks are measured, transmitted, and managed across interconnected financial systems.

Tail risk has become an essential lens for understanding financial markets in an era of heightened policy, geopolitical, and climatic uncertainty. Traditional risk measures (e.g. variance, VaR) struggle to capture the possibility of extreme losses, especially when shocks cascade across interconnected markets. When a major policy pivot or unexpected tariff shock occurs, it can amplify stress across equities, bonds, commodities, and alternative assets simultaneously, creating feedback loops and systemic fragility.

A growing body of research explores tail risk through diverse lenses: option-implied measures (Kelly and Jiang, 2014; Gao et al., 2019; Ilhan et al., 2021), return-based indicators (Bollerslev and Todorov, 2011; Christoffersen et al., 2012; Maheu et al., 2013), macroeco-

¹In conjunction with the tariff announcement, the U.S. introduced export controls on critical software and signaled imposition of port fees (50 USD per net ton) on Chinese-built vessels. In retaliation, China unveiled special port fees on U.S. ships (starting at 400 CNY per net ton), initiated an antitrust investigation into Qualcomm, and defended its strengthened export controls on rare earth materials and associated technologies, citing national security concerns.

nomic disaster models (Schreindorfer, 2020; Loria et al., 2025), and more recently, Gen-AI based simulations (Cont et al., 2025). These studies highlight the predictive and pricing power of tail risk but remain fragmented across asset classes, markets, and methodologies.

In this landscape, a systematic exploration of how tail risk is measured, transmitted, and managed is overdue. By surveying the diverse approaches: option-implied metrics, extreme value theory, systemic network models and mapping their applications across asset classes and geographies, this review aims to bring coherence to a fragmented field. Ultimately, understanding tail risk is not just academic: it is central to designing resilient financial systems capable of withstanding the next shock.

In the remainder of this pre-registered report, Section 2 presents the research question, Section 3 outlines the literature foundations, Section 4 details the data sources, tools, and methodological protocol, Section 5 reports our planned output, and Section 6 concludes.

2 Research Questions

2.1 Background and Research Context

Over the past two decades, financial markets have experienced a series of low probability but huge-impact shocks that have fundamentally reshaped perceptions of risks. The 2008 Great Financial Crisis (GFC) did not only increase volatility, it also revealed how fast losses can amass when leverage and connected risk exposures unwind together. The COVID-19 pandemic in 2019 exhibited a different kind of shock, where liquidity and risk preference vanished almost overnight after announcement. More recently, geopolitical conflicts and policy surprises have repeatedly produced abrupt repricing across multiple markets. These scenarios point to a recurring pattern: the most damaging outcomes sit in the far 'Left Tail', and they always arrive with little or no early-warning signs.

Tail-risk research has expanded rapidly in multiple directions, spanning option-implied

measures, return-based indicators, macro-disaster frameworks, and more recently, AI-driven (and generative AI, GenAI) approaches. However, it remains difficult to compare findings because studies often rely on different definitions, proxies, and market settings. As a result, evidence on how tail risk is measured, priced, transmitted, and mitigated is scattered across the literature. This fragmentation makes a systematic literature review necessary to identify current research themes, reconcile competing measures, and highlight future research gaps.

2.2 Research Questions

This systematic literature review (SLR) seeks to provide a comprehensive synthesis of the academic literature on tail risk in financial markets. While tail risk has been extensively studied in recent years, the research landscape remains highly fragmented, with studies employing different methodologies, focusing on varied asset classes, and addressing distinct market contexts. Moreover, theoretical and practical dimensions such as the underlying sources of tail risk, its causal mechanisms, and strategies for risk mitigation have yet to be systematically integrated into a coherent framework.

Accordingly, our guiding research question is:

”What are the key themes, empirical findings, theoretical perspectives, and future research directions in studies on tail risk in financial markets?”

To answer this question, the review will critically evaluate existing research along several interconnected dimensions including measurement methodologies (such as option-implied, return-based, and macroeconomic approaches), asset classes (equities, fixed income, commodities, and alternative assets), and market contexts (developed and emerging). The review will also examine underlying risk sources and causal mechanisms, the role of theoretical frameworks such as rare-disaster and behavioral finance models, and the evolution of tail-risk management and hedging strategies.

By synthesizing these strands, the SLR aims to develop a unified conceptual frame-

work that links the measurement of tail risk to its market pricing, systemic transmission, and mitigation. This framework will provide a structured foundation for understanding extreme financial events and offer a coherent research agenda to guide future empirical and theoretical advances.

3 Literature Foundations and Motivation

3.1 Three Key Reference Papers

Our Phase III pre-registered report is guided by the Phase II pitch table (Table A in the Appendix). Three seminal studies form the conceptual and methodological foundation for this review. Together, they illustrate the evolution of tail-risk research from market-based measurement to macroeconomic interpretation.

Our first key paper, [Kelly and Jiang \(2014\)](#), propose an innovative cross-sectional measure of tail risk extracted from firm-level return distributions. Their “Tail Beta” captures extreme downside co-movements and predicts both aggregate equity returns and real economic activity. This work is pivotal because it translates the abstract notion of rare disaster risk into an observable market statistics, providing the first scalable empirical proxy for economy wide tail exposure. It anchors the micro-to-macro transmission channel that later studies extend through option-implied and network-based frameworks.

Our second key paper, [Bollerslev et al. \(2015\)](#), decompose the variance risk premium into its diffusive and jump components and demonstrate that the jump-tail component dominates return predictability. Their non-parametric approach, using short-maturity OTM (out-of-the-money) option prices, isolates risk-neutral tail probabilities and links them to physical-measure expectations. This study consolidates the option-implied strand of the literature and provides the methodological template for measuring priced tail-risk premia across asset classes.

Our third key paper, [Marfe and Pénasse \(2024\)](#), shift the focus from financial markets to macroeconomic fundamentals. By estimating time-varying probabilities of rare disasters across 42 countries using quantile regressions of consumption growth, they construct a macro tail-risk index that predicts global equity returns. Their results bridge the asset-pricing and macro-risk literature, highlighting that disaster risk is a dynamic and measurable macro factor rather than a purely exogenous shock.

Together, these three pivotal studies capture the evolution of tail risk research across complementary dimensions, spanning micro- and macro-level analyses, return-based and option-implied methodologies, and market- and economy-wide perspectives. Yet, despite their collective influence, no existing study has systematically compared these approaches or synthesized their insights across asset classes, market contexts, and theoretical foundations. This gap motivates our proposed SLR, which aims to map, classify, and evaluate the literature to reveal consistent patterns, methodological divergences, and unaddressed research opportunities in the study of tail risk.

3.2 Boarder literature

3.2.1 Measurement of Tail Risk

Tail risk measurement faces a dilemma: the outcomes of tail risk are rare, and its inference must be drawn from limited observations ([Bee et al., 2016](#); [Daouia et al., 2021](#)). This has led to several tail measurement that often operate in parallel. One measure that use option prices, especially those deep out-of-the-money (OTM) contracts, to infer forward-looking crash insurance and recover the left-tail shape under the risk-neutral measure, often emphasizing the role of jump-related downside risk ([Bates, 1991](#); [Andersen et al., 2017](#); [Chen et al., 2019](#); [Fan et al., 2022](#)). Another measure relies on realized returns, estimating tail dependence, tail betas, or tail factors from extreme co-movements in the data ([Harris et al., 2019](#); [Bollerslev et al., 2022](#)). The last measure estimates tail risk from

macroeconomic fundamentals, modeling the lower tail of consumption or output growth and connecting these downside risks to financial variables (Segal et al., 2015; Marfe and Pénasse, 2024).

Methodological work further highlights that these measures can behave differently depending on sampling frequency (Bee et al., 2016), dependence structure (Stoja et al., 2023), and model risk (Stoja et al., 2025), making validation and comparability nontrivial. For this reason, our proposed SLR treats measurement methodology as a core classification dimension and will document, for each study, what object is being measured (risk-neutral v.s. physical, single-asset v.s. systemic, market-based v.s. macro-based) and the assumptions required to interpret it.

3.2.2 Pricing of Tail Risk

Tail risk is a statistical feature of asset return distributions, it also been identified by a large pricing literature as an economically meaningful risk that commands compensation in asset markets (Kelly and Jiang, 2014; Hu and Sun, 2021; Fan et al., 2022). Investors are willing to pay for a high crash insurance, so assets that exposed on extreme downside states (Tail events) tend to earn different expected returns or exhibit return predictability linked to tail-related risk premia (Constantinides et al., 2013; Christoffersen et al., 2017).

Empirically, such pricing mechanism in tail risk is tested in the following two ways. Firstly, cross sectional studies inspect whether assets with strong tail exposure exhibit different expect returns compared with other assets (Aboura and Arisoy, 2019; Chen et al., 2019; Jiang et al., 2025). Secondly, time series studies examine whether the variation of tail risk premia could predict future market returns (Suh et al., 2021; Marfe and Pénasse, 2024). Furthermore, the results are not always constant across various asset classes, markets and tail-risk measures. These patterns inspired our proposed SLR to inspect tail-risk pricing papers by measures, asset classes, stress regimes and market settings.

3.2.3 Transmission Mechanisms and Drivers of Tail Risk

Research find that tail risk is not only existed in single asset or market, instead, it often transmitted across financial markets, especially when liquidity is low or exhausted. Studies find spillovers and co-tail risk dependence across markets, where margin/short-selling constraints and fire-sales/deleveraging transmit extreme stress from one market/sector to another, which turning local shocks into tail events (Fostel and Geanakoplos, 2008; Falato et al., 2021; Huang et al., 2023). Network and connectedness formalize this mechanism by showing how balance-sheet exposures and trading relationships can cluster tail risk, especially during those crisis periods (Diebold and Yilmaz, 2014; Acemoglu et al., 2015; Glasserman and Young, 2016; Duarte and Eisenbach, 2021).

On the driver side of tail risk, research focused on the sources of tail vulnerability, which include jump risk, disaster concerns, leverage and liquidity constraints and investors' risk preference (Brunnermeier and Pedersen, 2009; Bollerslev and Todorov, 2011; Bekaert et al., 2013; Segal et al., 2015; Huang et al., 2023). Furthermore, other research links tail-events to information frictions (Hutton et al., 2009), investor sentiment/attention (Stambaugh et al., 2012), policy shocks (Bekaert et al., 2013), geopolitical conflicts (Caldara and Iacoviello, 2022) that trigger the repricing of tail risk. These findings suggest a distinction between the drivers and transmission mechanisms, which our proposed SLR will code separately to synthesis tail risk literature across different assets and markets.

3.2.4 Tail Risk Management and Emerging Methods in Tail Risk measurement

Once tail risks are identified by scholars and practitioners, the next natural question is how investors and institutions could manage such risks? Numerous studies classify tail risk hedging as crash insurance often through instruments that designed to pay only when markets crashed (e.g., Bates, 1991; Barro, 2006; Carr and Wu, 2009; Bollerslev and Todorov, 2011; Kelly and Jiang, 2014; Andersen et al., 2017; Doshi et al., 2025). One of com-

mon method is long deep out-of-money put options and volatility-linked products, which could provide 'insurance' for investors in markets crashes (Carr and Wu, 2009; Bollerslev and Todorov, 2011; Moreira and Muir, 2017; Suh et al., 2021). Other research focused on portfolio and strategies, emphasizing on how dynamic re-balancing, volatility management/timing, or robust portfolio rules to hedge extreme downside risk (Ang et al., 2006; Fleming et al., 2001; Asness et al., 2012; Kelly et al., 2019). This literature matters as it raised a uncomfortable trade-off for investors: tail risk protection is not free, and the effectiveness of hedge depending on various factors (e.g., liquidity, market regime, risk preference).

More recently, some papers has begun to use data-driven simulation to investigate tail risks instead of tradition parametric models, these studies implement new methodology like machine learning to generate paths to simulate extreme events (e.g., Gu et al., 2020; Bali et al., 2023; Delle Monache et al., 2024). As history only offer a limited number of tail events/crises (e.g., 2008 GFC, COVID-19 pandemic), simulation can extend potential tail events, especially for multi-asset portfolios where asset-co-movement matters. Meanwhile, these new methods had introduced concerns for interpret-ability, stability, and whether 'simulation' could reflect genuine event or simply new method is a kind of data mining (Ang and Bekaert, 2002; Buehler et al., 2019; Kelly et al., 2019; Giglio et al., 2021). For this facts, our proposed SLR will classify tail-risk management and emerging simulation methods as a field to coding academic queries by (a) the hedging instrument used, (b) the target tail concept, and (c) whether the approach is model-based, data-driven, or simulation.

3.3 Motivation

Based on our brief analysis of related literature and phase 2 pitch table. We find that tail risk is crucial to understanding asset pricing anomalies in financial markets, especially, how

the financial system behave during crisis. However, the evidence is scattered across various tail measurement: option-implied metrics, return-based metrics, and macro-fundamental indicators, and often studies focused within certain asset classes and markets (e.g., [Constantinides et al., 2013](#); [Bollerslev et al., 2015](#); [Hu and Sun, 2021](#); [Marfe and Pénasse, 2024](#)). This fragmentation motivates a structured, PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)-guided systematic literature review that integrates finding across methodologies and settings, that could clarify what different proxies capture, and identifies robust research gap and future research direction on financial tail risk.

4 Data, Tools and Methodology

4.1 Data Sources

This SLR will adopt a series of transparent protocol to identify, screen and synthesize research on tail risk in financial market. We primarily rely on literature meta-data from two of the most widely used academic database by literature review papers: Elsevier Scopus and Clarivate Web of Science (WoS) (e.g., [Nazareth and Reddy, 2023](#); [Kou et al., 2024](#); [Buchetti et al., 2025](#)).² We will build Boolean search strings aligned with our classification to extract literature data from these two databases.

Meanwhile, to reduce omission risk and include frontier work, we will also incorporate highly-cited working papers (grey literature) from working paper repository hub: NBER and SSRN, which will be analyzed separately from peer-reviewed published papers. In addition, given the growing popularity of generative-AI (GenAI) and simulation-based techniques in tail risk modeling. The SLR will also screen highly-cited relevant studies from leading machine learning academic conferences (e.g., NeurIPS, ICML, ICLR) when they

²NBER (National Bureau of Economic Research) and SSRN (Social Science Research Network) are widely used platforms for disseminating finance and economics unpublished working papers prior to journal publication.

are directly related to tail-risk measurement.³

Our primary synthesis will focus on peer-reviewed papers published in leading finance, economics, and interdisciplinary journals (e.g., SSCI-indexed and journals listed in ABDC/ABS rankings), to ensure the review reflects rigorous and influential contributions. All exported records will be stored in reference-management software (Zotero) for de-duplication and screening, and the selection process will be clearly documented using a PRISMA flow diagram in the full study.

4.2 Search strategy

In line with Phase III pre-registration guidelines ([Faff, 2023](#)), our pre-registered report specifies the review design and procedures, and does not report completed findings. Following the data sources specified above, we will implement Boolean searches in Scopus and Web of Science (with supplementary searches in NBER/SSRN and targeted screening of NeurIPS/ICML/ICLR), applying a keyword system aligned with our review framework.

We will implement the search strings to the 'Title', 'Abstract' and 'Keywords' in WoS and Scopus. Consistent with other SLRs, languages will be restricted to English-language in the window. Where database filters allow, we will limit our search results within the subject to Finance, Economics, and closely related interdisciplinary subjects (for example, Social Science, Business). Working papers and conference papers will be coded separately as the repository portals usually do not provide language or subject filters.

Our searching keywords is aligned with our review framework. Each query includes at least one core tail risk term (e.g., 'tail risk', 'crash risk', 'downside risk') with at least one dimension term that capture areas in board literature: measurement, pricing, transmission,

³NeurIPS (Conference on Neural Information Processing Systems), ICML (International Conference on Machine Learning), and ICLR (International Conference on Learning Representations) are leading peer-reviewed academic conferences in machine learning and AI, where many methodological advances are first disseminated.

drivers, management and merging method. This combination design could reduce omission risk from difference in terminology across difference sub-fields.

Based on the search logic, we will use Boolean query operator ('AND'/'OR'), wildcards ('*') and phrase matching to implement replicable Boolean search queries. The query will be constructed through three query families that consistent with our board-literature: (i) measurement and pricing, (ii) transmission mechanisms and drivers, and (iii) management and emerging methods. Planned representative Boolean query strings include:

- (1) (“tail risk” OR “crash risk” OR “downside risk”) AND (“option-implied” OR “variance risk premium” OR “jump risk” OR “expected shortfall”),
- (2) (“tail risk” OR “systemic tail risk”) AND (“contagion” OR “spillover” OR “connectedness” OR “fire sale” OR “funding liquidity”),
- (3) (“tail risk” OR “crash insurance”) AND (“tail hedge*” OR “OTM put” OR “variance swap” OR “scenario generation” OR “generative model”).

The complete list of finalized strings in addition to the planned representative strings will be reported in the Phase IV full study.

To enhance completeness of the search query, we will implement backward and forward citation search from key papers identified in Phase II for highly-cited measurement papers. Additionally, we will also review the keywords of included studies to identify alternative terminology words (e.g., 'crash insurance', 'rare risk' and 'disaster risk') and confirm that these terms are included by the search queries.

4.3 Inclusion and Exclusion criteria

Based on our retrieved data from Boolean query, we will implement inclusion and exclusion criteria refine our final dataset.

4.3.1 Inclusion criteria

Research papers will be included if they meeting all of the following standards:

- (1) The studies must be related to tail risk in financial markets, which include one of the following: tail-risk measurement, tail-risk pricing/premia, systemic tail-risk transmission/contagion, drivers/causes of tail risk, tail-risk management/hedging, or AI/simulation-based tail-risk modeling.
- (2) The studies must be written in English and can be accessed from academic databases specified above.
- (3) The studies must be published in a peer-reviewed outlet (journal papers) or an influential repository (working papers).
- (4) Selected papers must provide a clear methodological approach that sufficient for coding under our review framework.

4.3.2 Exclusion criteria

Research papers will be excluded if they meeting any of the following standards:

- (1) The studies that use 'tail' terminology in a non-financial sense (e.g., statistical or mathematical tails in non-financial market setting).
- (2) The studies that purely descriptive about market commentary, news articles, textbooks, and non-academic reports.
- (3) The studies that focusing on extreme risk in domains that not related to financial markets (e.g., engineering, medical/health-care engineering tail risks).
- (4) The studies that without full-text access on academic databases (e.g., subscription of ScienceDirect, Wiley, Taylor & Francis).

- (5) The studies that without sufficient methodological details.
- (6) Duplicated studies emerged from merging WoS and Scopus datasets during de-duplication.

Accordingly, these criteria are designed to ensure the SLR only include relevant and high-quality studies on tail risk while maintaining the replicable PRISMA framework.

4.4 Screening Procedure

Following the search strategy reported above, all the retrieved records will be exported into reference management software (Zotero) and merged into a master library. We will use its automatic matching (title/authors/DOI) and manual check for near-duplicate papers. Then the screen process will be implemented in the following stages:

- (1) Stage 1: Title-Abstract screening, we will remove those irrelevant studies that do not focus on tail risk in financial markets or do not meet the inclusion criteria.
- (2) Stage 2: Full-text screening, we will evaluate the remaining studies' full-text against the full inclusion/exclusion criteria to confirm topic relevance and methodological details to be coded under our framework.
- (3) Stage 3: Final eligibility, we will generate the final literature dataset and document reasons for excluded papers.

To enhance transparency in the review, the screen process will be documented in the PRISMA flow diagram. We will maintain a screen log that record reason for exclusion. Working papers and conference papers will be clearly flagged in the log rather than being pooled together with journal papers.

4.5 Data Extraction and Coding Scheme

After the filtration of the final literature dataset, we will extract information from each paper using a standard coding form to ensure the SLR's replicability. The extraction data include bibliographic metadata (authors, year, type), research context (asset class) and the core analytical content. For transparency, all coding fields and definitions will be compiled into a code book attached in the appendix section.

The coding scheme is lined with our unified conceptual framework and organizes each study along key dimensions. They will be implemented in the following steps illustrate in Figure 1.

[Please Insert Figure 1 About Here]

In addition to the thematic coding reported above, the SLR will extract brief methodological description to support the assessment and comparison across studies: sample period, country/region, data frequency, identification strategy (descriptive, predictive, quasi-experimental, structural, simulation), and key outcomes. These supplementary extractions will facilitate the development of a summary table and the mapping of frameworks in the final full SLR study.

4.6 Planned Synthesis Method

Following the data extraction and coding framework of SLR, we will synthesize the included studies using a structured approach guided by our unified conceptual framework. The synthesis will be organized around three integrated strands we have previously identified: (i) measurement and pricing of tail risk, (ii) transmission mechanisms and drivers of tail risk, and (iii) tail risk management and emerging methods. Within each strand, we will assess each paper and group them into consist families of tail risk measures and classify them by asset classes and market setting (developed and emerging).

For transparency, the synthesis will combine descriptive mapping and comparative thematic synthesis. The descriptive mapping will report the distribution of research across doing dimensions, and will be represented by summary tables and framework based figures. The comparative synthesis will assess convergent and divergent findings by tracking the difference from (i) tail measurement choices, (ii) data (frequency, sample period, region/country), and (iii) identification strategies (descriptive, predictive, quasi-experimental, structural, or simulation-based). We will document the conditions under which results differ and evaluate the causes of these divergences.

The final SLR will include several synthesis artifacts to improve transparency and replicability: (i) a summary table that mapping tail risk measures to the objects they capture and their typical assumptions; (ii) a tabulation of themes by market types and asset classes; (iii) a mechanism table linking proposed drivers; and (iv) a dedicated synthesis of tail risk management and hedging studies, instruments from strategies and effectiveness. Studies using emerging GenAI or simulation-based tools will be coded and discussed as a distinct methodological frontier in comparison with tradition approaches.

5 Planned Outputs

Based on our proposed SLR review plan, the full SLR will produce a defined set of outputs that translate the proposed SLR into transparent full study. Figure 2 provides an overview of our planned deliverables, which include the PRISMA documentation, publication landscape summaries, core classification and taxonomy tables, cross-market theme summaries, a mechanism map linking drivers to transmission channels, a dedicated synthesis of tail-risk management and hedging studies, and a research-agenda map. Bibliometric visualizations (e.g., VOSviewer co-citation or co-occurrence networks) will be generated where metadata coverage is sufficient.

[Please Insert Figure 2 About Here]

6 Conclusion

6.1 What's new?

This pre-registered SLR offers a planned structured synthesis of tail-risk research that is currently dispersed in multiple strands. Rather than focusing on a single tail-risk proxy or market, the review is organized around a unified framework linking tail-risk measurement, pricing, transmission, drivers, and mitigation, while also incorporating emerging GenAI-based simulation approaches where relevant. The review protocol is specified with a PRISMA-guided search and screen procedures with a transparent code book to support replicability.

6.2 So What?

A clear review of tail risk literature is timely as tail events remain central to financial market stability. By investigate what different tail-risk measure capture, their assumption, and how evidence varies across assets and markets. The SLR is intended to provide academicians and investors a clear comparison in tail-risk literature. For policymakers, the synthesis can improve the interpretation of tail-risk indicators and identify which mechanisms and hedging approaches most relevant under different scenarios.

6.3 Contribution

The planned SLR contributes by investigate a fast growing literature into integrated analysis of synthesis (classification tables, measure families, mechanism maps, and a structured research agenda). This planned SLR also bridges theory, empirical and risk management practice by integrating tail-risk drivers, transmission channels, instruments, and risk management. Lastly, by documenting searching strings, screening procedures, and coding

rules, this planned SLR provides a replicable foundation that can be updated to related studies.

7 Appendix

Table [A](#) presents our Phase II pitch table, which serves as the guiding framework for the present Phase III pre-registered report.

[Please Insert Table [A](#) About Here]

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used [Sider AI/Writefull] in order to [proofread and improve the manuscript's readability]. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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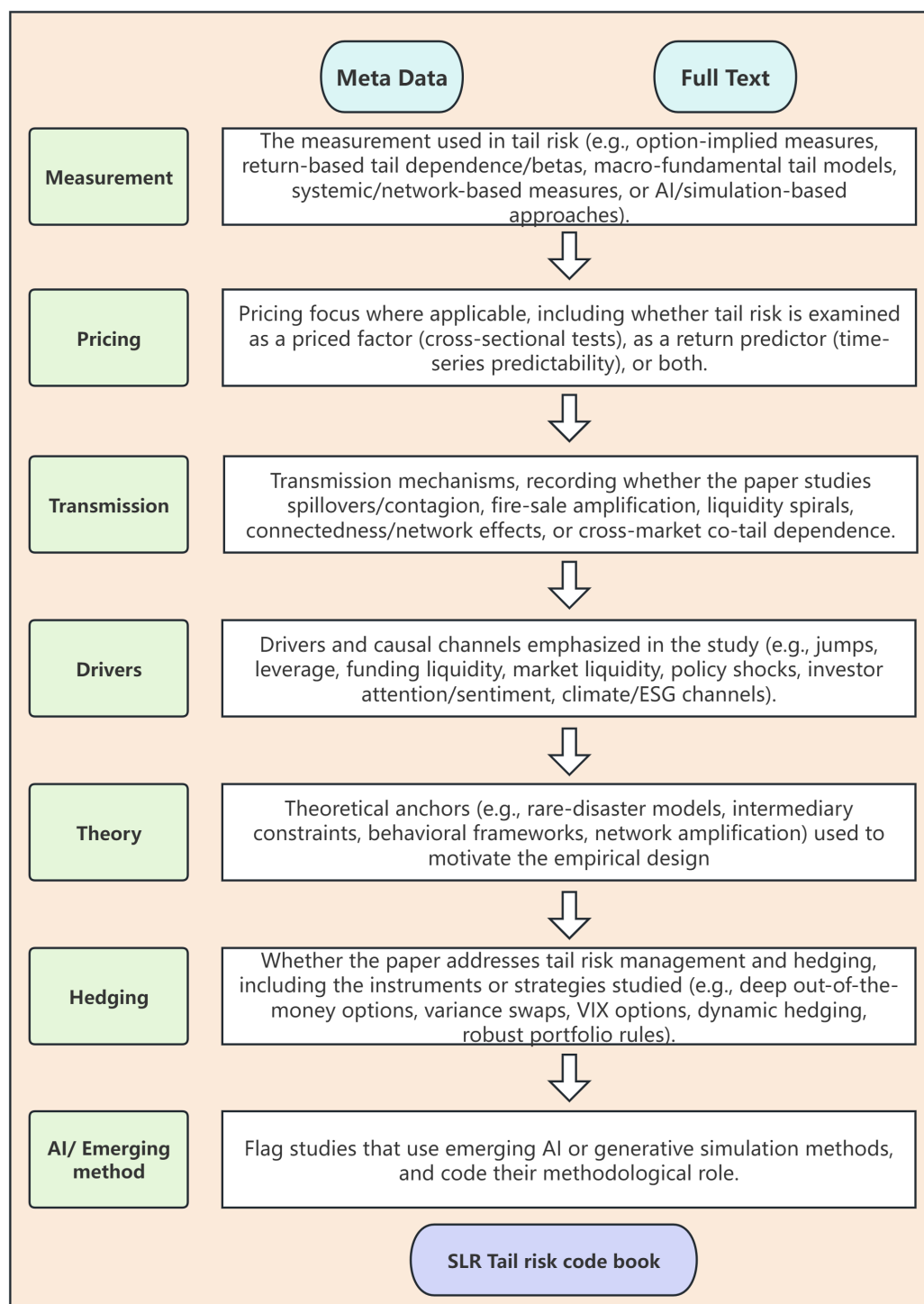


Figure 1: Code book and coding workflow for data extraction and classification
The diagram outlines the code book fields and coding flow used for systematic extraction and classification.

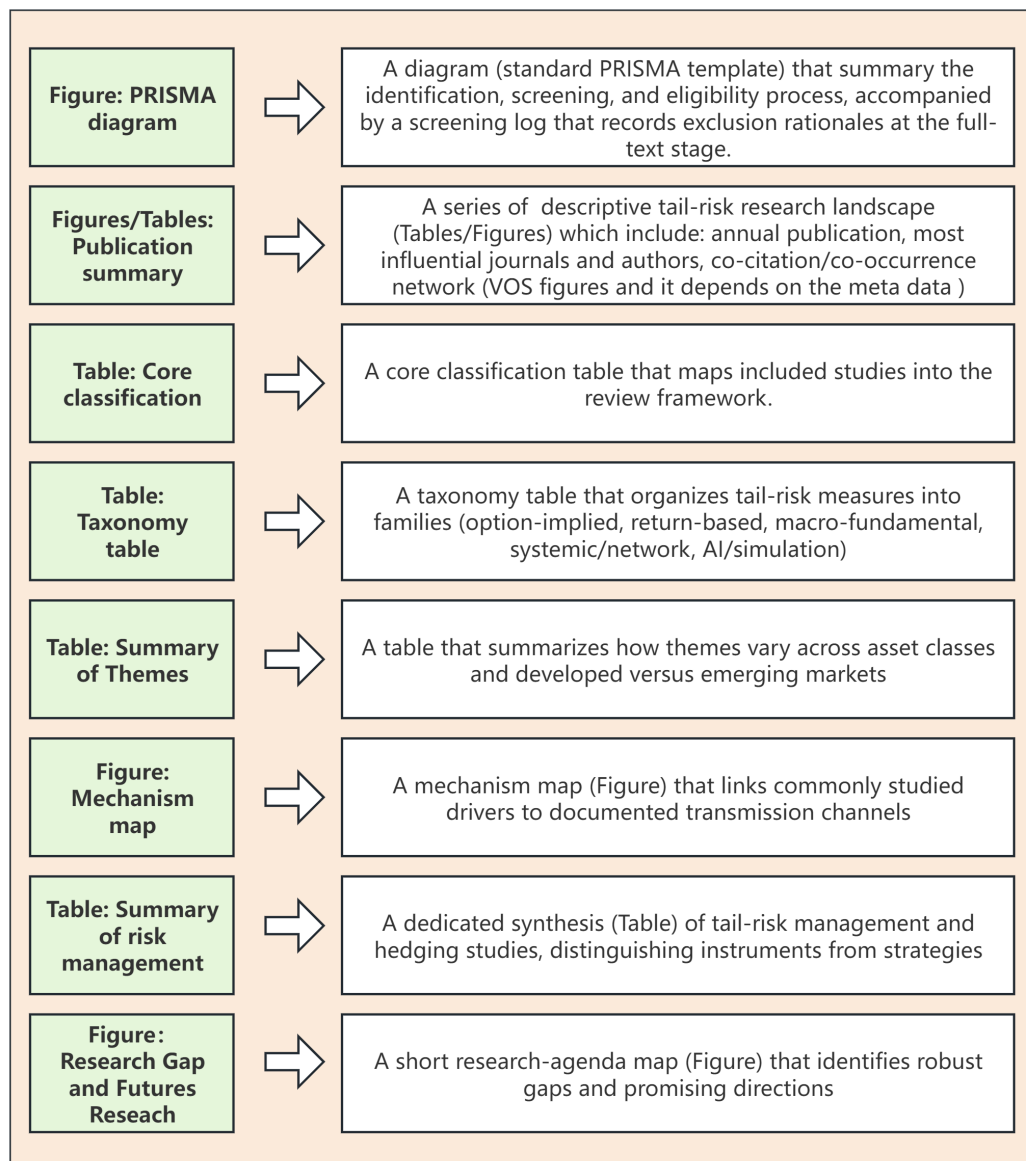


Figure 2: Planned outputs of the systematic literature review.

This figure summarizes the planned tables, figures, and bibliometric outputs to be produced in the full SLR.

Table A: Pre-registered pitch table

Pitcher Team Names	Author A, University Author B, University Author C, University
(A) Working Title	Systematic Literature Review of Tail Risk in Financial Markets
(B) Basic Research Question	What are the key themes, empirical findings, and future research directions in studies on tail risk in financial markets?
(C) Key papers	(1) Kelly, B., & Jiang, H. (2014). Tail Risk and Asset Prices. <i>Review of Financial Studies</i>, 27(10), 2841–2871. DOI: 10.1093/rfs/hhu039 • Why? This paper introduces a novel empirical measure of tail risk based on the left-tail covariance of firm-level equity returns, and shows its significant predictive power for aggregate market returns and macroeconomic activity. • How? Using a large panel of U.S. stocks, the authors extract a time-varying tail risk factor that captures extreme downside co-movements, which is then linked to future equity premia and economic contractions. • Contributions: This is a foundational paper in measuring tail risk from realized equity returns. It is essential for understanding how firm-level crash sensitivities aggregate into market-level systemic risk. (2) Bollerslev, T., Todorov, V., & Xu, L. (2015). Tail Risk Premia and Return Predictability. <i>Journal of Financial Economics</i>, 118(1), 113–134. DOI: https://doi.org/10.1016/j.jfineco.2015.02.010 • Why? This paper offers a pivotal contribution by decomposing the variance risk premium into distinct components and showing that the portion associated with jump tail risk is the primary driver of return predictability. • How? The authors develop a nonparametric method using short-maturity, deep out-of-the-money (OTM) options to estimate risk-neutral jump tail shapes, enabling them to isolate and track time-varying tail risk premia linked to market fears. • Contributions: This study investigates how option-implied tail risk premia influence asset returns, providing a methodological benchmark and empirical evidence critical for options-based tail risk analysis. (3) Marfè, R., & Pénasse, J. (2024). Measuring Macroeconomic Tail Risk. <i>Journal of Financial Economics</i>, 156, 103838. DOI: https://doi.org/10.1016/j.jfineco.2024.103838 • Why? This paper addresses a key gap in the literature by estimating macroeconomic tail risk directly from economic fundamentals, rather than relying on asset prices, thereby allowing for a cleaner test of rare disaster theories. • How? It employs a semi-parametric two-step approach using quantile regressions on a broad set of predictors across 42 countries to estimate the probability of extreme downturns in consumption, constructing a novel macro risk index. • Contributions: This paper adds a macroeconomic dimension to tail risk measurement and connects it to asset pricing outcomes. It is particularly valuable for understanding the time-varying nature of disaster risk and its empirical implications across countries.
(D) Motivation/Puzzle:	(1) Tail risk plays a critical role in explaining asset pricing anomalies and systemic market behavior under stress. (2) Existing studies span diverse methods (e.g., options, returns, macro data) but remain fragmented. (3) A structured SLR review is needed to integrate findings, expose gaps, and guide future research on financial tail risk.
THREE	Three core aspects of any empirical research project

Table A: Pre-registered pitch table (continued)

(E) Idea?	This SLR focuses on tail risk in financial markets, emphasizing its measurement, pricing, and transmission. (1) We organize the literature into the following key dimensions: (a) Methodology (e.g., option-implied measures), (b) Asset classes (e.g., equities, fixed income, commodities), (c) Markets (e.g., developed vs. emerging), (d) Risk source (e.g., macroeconomic shocks, financial contagion, policy shifts), (e) Drivers (e.g., jump risk, volatility clustering, liquidity conditions), (f) Theoretical frameworks (e.g., structural models, behavioral finance), (g) Tail risk management and hedging strategies (e.g., deep out-of-the-money, OTM options, variance swaps, and dynamic hedging models). (2) We also examine key drivers of tail risk, such as variance risk premia, jump risk, and market liquidity. (3) By synthesizing diverse approaches and contexts, we aim to provide a unified conceptual framework for understanding financial tail risk. An integrative structure that links diverse tail risk measurement methodologies (option-implied, return-based, and macroeconomic), maps their transmission channels across asset classes and markets, identifies underlying risk sources and causal drivers, and connects these insights to practical tail risk management and hedging strategies.
(F) Data?	The SLR will follow a structured and transparent approach to identify, screen, and synthesize scholarly research related to tail risk in financial markets. To ensure broad and high-quality coverage, we will source literature from multiple academic databases such as Elsevier Scopus and Clarivate Web of Science (WoS). Our institutional access to these databases provides comprehensive availability of peer-reviewed journal articles, working papers, and relevant book chapters. The review will prioritize peer-reviewed studies published in leading finance, economics, and interdisciplinary journals (e.g., SSCI, ABDC, and ABS-indexed), ensuring that the analysis reflects the most influential and rigorous contributions in the field.
(G) Tools?	(1) This SLR adopts a structured, replicable approach to synthesize a fragmented literature on tail risk in financial markets, minimizing bias and enhancing transparency. (2) We follow the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework for search, screening, and inclusion procedures. (3) Academic databases used include Scopus, WOS, and other leading repositories, with Boolean search operators applied to refine results using keywords. In addition to peer-reviewed papers, we will include influential working papers from reputable platforms such as NBER and SSRN. The keyword system will be comprehensive, covering terms related to tail risk measurement (e.g., option-implied, return-based, macroeconomic indicators), transmission mechanisms, risk sources, hedging strategies, and theoretical frameworks. Beyond article selection, we will synthesize findings through thematic coding along the dimensions of methodology, asset class, market, risk source, theory, and management strategy, ensuring transparency and replicability in the synthesis stage. (4) Given the increasing rising of AI, the search will also include potentially relevant studies from leading AI and machine learning conferences (e.g., ICML, ICLR, NeurIPS) to capture emerging generative-AI-based methods for modeling complex market and systemic risk dynamics. Relevant key words related to AI will be included (e.g., Generative AI, Financial systemic risk AI). (5) Inclusion criteria focus on peer-reviewed studies in English, primarily from high-impact finance and economics journals. (6) Reference management tools (Zotero) will be employed to organize and track selected studies throughout the review process.
TWO	Two key questions
(H) What's New?	(1) This is the first SLR to comprehensively map the academic landscape of tail risk in financial markets across methodologies, asset classes, and geographic regions (To the best of our knowledge and our corresponding search in Google Scholar, Scopus and WOS). (2) Unlike previous reviews that are typically narrow in scope, focusing on a single asset class, market, or methodology and often lack integration between theory, empirical evidence, and practical considerations such as management and hedging, our study systematically integrates theoretical frameworks with empirical evidence, offering a more comprehensive understanding of tail risk drivers and implications. Our review overcomes these limitations by unifying diverse methodologies, asset classes, markets, and theoretical perspectives into a single, coherent conceptual framework. (3) We integrate insights from equity, fixed income, and commodity markets, which spanning both developed and emerging economies to identify patterns, contradictions, and gaps in the literature.

Table A: Pre-registered pitch table (continued)

(H) What's New?	(4) Our review uniquely integrates a practical perspective on tail risk management and hedging, an area largely absent from existing literature reviews, providing actionable insights for portfolio construction and risk mitigation. (5) The review contributes a unified conceptual framework, an integrative structure that connects diverse tail risk measurement methodologies (option-implied, return-based, and macroeconomic approaches), maps their transmission mechanisms across asset classes and markets, identifies underlying risk sources and causal drivers, and links these to practical tail risk management and hedging strategies.
(I) So What?	This study conducts a systematic literature review (SLR) of tail risk in financial markets, focusing on its measurement, pricing, and transmission. It aims to synthesize fragmented research across methodologies (e.g., option-implied vs. return-based), asset classes (equities, fixed income, commodities), and markets (developed and emerging) to provide an integrated understanding of tail risk.
ONE	One bottom line
(J) Contribution?	This SLR offers the first structured synthesis of tail risk research across financial markets (To the best of our knowledge). This framework integrates measurement methodologies, transmission mechanisms, risk sources, and mitigation strategies, providing a structured lens to interpret diverse findings in a coherent way. It clarifies core themes, identifies methodological gaps, and organizes findings across asset classes and regions. The review bridges the gap between theory and empirical research by organizing literature according to both methodological approaches and underlying theoretical frameworks. The review connects academic research with practical strategies for tail risk mitigation, offering insights for effective risk control in extreme market conditions. The review provides a foundation for future empirical and theoretical work by proposing a unified framework for understanding the role of tail risk in asset pricing and financial stability.
(K) Other Considerations	The intended organization: The final manuscript will be structured around the components of the unified framework, with sections on measurement, transmission, sources, theory, and management, ensuring a logical and coherent narrative.